A SUBMODULE OF A QUASI-COHERENT MODULE CONTAINS A LARGEST QUASI-COHERENT SUBMODULE, CHARACTERISED BY ITS UNIVERSAL PROPERTY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The canonical Stacks tag of the counted statement is 01QZ (morphisms-lemma-largest-quasi-coherent-subsheaf). Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the tag is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of sources/stacks-chow/preamble.tex lines 14-18 and 154-217. No macro was invented or redefined.

Editorial note. The statement is self-contained (F, G, G' and the universal property are all named in it), so no context span is attached; the two same-chapter lemmas that surround it in the source are separate claims and are not counted here.

Allowed standard prerequisites (editorial).

- Images and direct sums of maps of quasi-coherent O_X-modules are quasi-coherent. Schemes, Section labelled schemes-section-quasi-coherent (canonical Stacks tag 01LA); invoked at proof lines 465-468. Supplies the two closure properties that make the constructed image sheaf G' quasi-coherent, and again makes the image of H -> G -> F quasi-coherent.
- A quasi-coherent O_X-submodule of a quasi-coherent O_X-module is a kernel of a map of quasi-coherent modules, hence itself quasi-coherent, and the set of quasi-coherent submodules contained in a fixed submodule is a set. Background from the same chapter and from Schemes, Section schemes-section-quasi-coherent (canonical Stacks tag 01LA). Makes the indexing set A of all quasi-coherent submodules contained in G available so that the direct sum over A can be formed.

Source reference. Original section 01LA: sources/stacks-chow/tags.txt, source lines 2079–2079. Cross-chapter pointer invoked at proof lines 465-468 for the two closure properties (direct sums and images of maps of quasi-coherent sheaves) on which the construction rests; the construction and the universal property are carried out in full inside the counted span.

Lemma 0.1. *Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $\mathcal{G} \subset \mathcal{F}$ be a $\mathcal{O}_X$-submodule. There exists a unique quasi-coherent $\mathcal{O}_X$-submodule $\mathcal{G}' \subset \mathcal{G}$ with the following property: For every quasi-coherent $\mathcal{O}_X$-module $\mathcal{H}$ the map*

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{H}, \mathcal{G}') \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{H}, \mathcal{G})$$

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is bijective. In particular $\mathcal{G}'$ is the largest quasi-coherent $\mathcal{O}_X$-submodule of $\mathcal{F}$ contained in $\mathcal{G}$.

Proof. Let $\mathcal{G}_a$, $a \in A$ be the set of quasi-coherent $\mathcal{O}_X$-submodules contained in $\mathcal{G}$. Then the image $\mathcal{G}'$ of

$$\bigoplus\nolimits_{a \in A} \mathcal{G}_a \longrightarrow \mathcal{F}$$

is quasi-coherent as the image of a map of quasi-coherent sheaves on X is quasi-coherent and since a direct sum of quasi-coherent sheaves is quasi-coherent, see Schemes, Section 01LA. The module $\mathcal{G}'$ is contained in $\mathcal{G}$. Hence this is the largest quasi-coherent $\mathcal{O}_X$-module contained in $\mathcal{G}$.

To prove the formula, let $\mathcal{H}$ be a quasi-coherent $\mathcal{O}_X$-module and let $\alpha : \mathcal{H} \to \mathcal{G}$ be an $\mathcal{O}_X$-module map. The image of the composition $\mathcal{H} \to \mathcal{G} \to \mathcal{F}$ is quasi-coherent as the image of a map of quasi-coherent sheaves. Hence it is contained in $\mathcal{G}'$. Hence α factors through $\mathcal{G}'$ as desired. □